

Report

CSE351 Computer Networks

Fall, 2025, Programming Assignment #1

Due: September 21, 2025 (Sun), 23:59 KST

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I divided my implementation into several files to keep the code organized and clear to read.

In **client.c**, I wrote the logic for the client program. It takes command line arguments, reads input from standard input, and then constructs messages according to the protocol. Each message has a header, key, and data, which I send to the server reliably using helper functions for sending and receiving. I also made sure the client can split very large input into smaller chunks that still respect the protocol size limits. Finally, it receives the server's response and writes the output directly to standard output.

In **server.c**, I implemented the server side. It creates a listening socket, accepts multiple client connections, and uses forked child processes to handle them in parallel. For each client message, the server validates the header, reads the key and data, and applies the Vigenère cipher. The server then sends back a properly formatted response. I also added handling for child process cleanup using a signal handler, and I limited the number of concurrent clients.

The cipher logic itself is in **vigenere.c**. I implemented both encryption and decryption functions, making sure they work with both uppercase and lowercase letters, while leaving non-alphabetic characters unchanged. To make these functions reusable, I declared them in **vigenere.h**, which I included in the server.

To manage building everything, I created a **Makefile**. It defines how to compile the client and server programs, links the cipher code, and includes a clean target so that I can remove the compiled executables easily.

By structuring the project this way, I separated responsibilities clearly: the client for preparing and sending data, the server for handling communication and applying the cipher, and the cipher module for the actual encryption and decryption. This design makes the code easier to maintain and ensures that each part only focuses on its own task. All the provided tests passed with accordance to all the assignment requirements.